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Case 30-2025: An 82-Year-Old Woman with Abdominal Distention, Edema, and Pleural Effusion

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PRESENTATION OF CASE

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N Engl J Med 2025;393:1634-44.

DOI: 10.1056/NEJMcpc2412538

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CME



Dr. Leanna J. Travis (Medicine): An 82-year-old woman presented to the emergency department of this hospital because of worsening abdominal distention and pedal edema.

Three weeks before the current presentation, the patient traveled to South America to visit friends and family. At this time, she was in her usual state of health and able to independently tend to her daily needs. During her time in South America, abdominal bloating and a sensation of heaviness in the legs developed. She had reported no substantial changes to her diet while she was in South America, and she had no known sick contacts. The patient stayed in urban areas and had had no exposure to insects or animals. She returned to the United States 1 day before the current presentation.

The next day, worsening edema of the lower legs limited the patient's ability to dress herself, and the patient's daughter brought her to the emergency department of this hospital for evaluation. Physical examination was notable for diminished breath sounds and pitting edema of the legs that extended to the thighs. Imaging studies were obtained.

Dr. Cynthia L. Czawlytko: An upright posteroanterior radiograph of the chest (Fig. 1A) showed cephalization of the pulmonary vasculature, mild central peribronchovascular cuffing, and peripheral septal lines, without overt enlargement of the cardiac silhouette. Blunting of the costophrenic angles was observed that was more pronounced on the left side than on the right, which may indicate a small left pleural effusion and a trace right pleural effusion, findings consistent with pulmonary edema.

Subsequently, computed tomography of the abdomen and pelvis (Fig. 1B) was performed before and after the administration of intravenous contrast material. The liver had a nodular contour, a feature commonly associated with cirrhosis; however, this finding is not diagnostic in isolation. In addition, small-volume ascites was present. The portal veins were patent, and no evidence of additional signs of portal

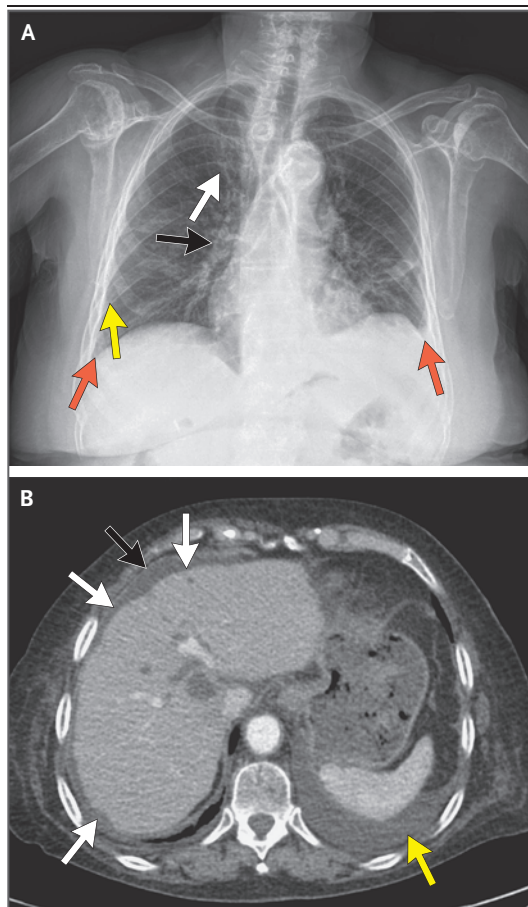


Figure 1. Imaging Studies of the Chest and Abdomen.

A posteroanterior radiograph of the chest (Panel A) shows cephalization of the pulmonary vasculature (white arrow), peribronchovascular cuffing (black arrow), and septal lines (yellow arrow). Trace right and small left pleural effusions are present (orange arrows). An axial CT image of the upper abdomen, obtained after the administration of intravenous contrast material (Panel B), shows a nodular contour of the liver (white arrows), small-volume perihepatic ascites (black arrow), and a small left pleural effusion (yellow arrow). The portal veins are patent. Hepatic cysts are incidental findings.

hypertension, such as splenomegaly or portosystemic shunting, was seen. Diffuse anasarca was noted throughout the visualized subcutaneous tissues.

Dr. Travis: Electrocardiography was performed and showed sinus rhythm. Treatment with intravenous furosemide was administered. The patient was discharged from the emergency department and was enrolled in the home hospital service for management of decompensated heart failure. The

home hospital service is an acute care delivery system in which inpatient-level care is administered in the home of qualifying patients.

On evaluation in the patient's home, she reported adherence to her medications during the trip to South America but had noticed less urine production than usual after taking her daily dose of furosemide. In addition, she reported a nonproductive cough that was worse at night, as well as orthopnea. A review of systems was negative for palpitations, chest discomfort, anorexia, syncope, easy bruising, night sweats, weight loss, rash, numbness and tingling, hemoptysis, and dizziness upon standing but was notable for frothy urine and hand and periorbital edema.

The patient's medical history included atrial fibrillation with cardioembolic stroke, for which she had undergone direct-current cardioversion and cardiac ablation through pulmonary-vein isolation 1 year before the current presentation, as well as heart failure with preserved ejection fraction. Medications included metoprolol, apixaban, and furosemide. She was born and raised in South America and had moved to suburban Massachusetts to live with her daughter and son-in-law 14 years earlier. She did not smoke, drink alcohol, or use recreational drugs. She had no known adverse reactions to medications and no family history of liver or kidney disease.

On examination, the oral temperature was 36.7°C, the blood pressure 114/77 mm Hg, the pulse 80 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 95% while the patient was breathing ambient air. The weight was 67 kg, and the body-mass index (the weight in kilograms divided by the square of the height in meters) was 30. The jugular venous pressure was 12 cm of water, and a low-pitched holosystolic murmur was heard at the left lower sternal border. The breath sounds were diminished at both lung bases without crackles or wheezing. Pitting edema was present throughout both legs and extended up to the lateral abdominal wall. The abdomen was distended with dullness on percussion. She had no bruising, Terry's nails (characterized by a white or pale nail bed with a narrow pink or brown band at the tip), spider angiomas, or palmar erythema. Urinalysis showed 2+ proteinuria with a specific gravity of 1.006, without hematuria or pyuria. Additional laboratory test results are shown in Table 1.

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Current Presentation
Hemoglobin (g/dl)	13.5–17.5	10.9
Hematocrit (%)	41.0–53.0	32.2
White-cell count (per μ l)	4500–11,000	3600
Differential count (per μ l)		
Neutrophils	1800–7700	2350
Lymphocytes	1000–4800	770
Monocytes	200–1200	250
Eosinophils	0–900	70
Basophils	0–300	20
Immature granulocytes	0–100	0
Platelet count (per μ l)	150,000–400,000	248,000
Sodium (mmol/liter)	135–145	141
Potassium (mmol/liter)	3.4–5.0	3.7
Chloride (mmol/liter)	98–108	108
Carbon dioxide (mmol/liter)	23–32	25
Urea nitrogen (mg/dl)	8–25	17
Creatinine (mg/dl)	0.60–1.50	0.60
Calcium (mg/dl)	8.5–10.5	7.8
Glucose (mg/dl)	70–110	101
Aspartate aminotransferase (U/liter)	10–40	35
Alanine aminotransferase (U/liter)	10–55	15
Alkaline phosphatase (U/liter)	15–115	217
Total bilirubin (mg/dl)	0.0–1.2	0.2
Albumin (g/dl)	3.3–5.0	1.7
Globulin (g/dl)	1.9–4.1	3.1
High-sensitivity troponin T (ng/liter)	0–9	30
Total cholesterol (mg/dl)	<200	152
Low-density lipoprotein cholesterol (mg/dl)	50–129	77
Prothrombin time (sec)	9.4–12.5	12.0
Activated partial-thromboplastin time (sec)	22.0–36.6	36.0
International normalized ratio	0.8–1.1	1.0
Thyrotropin (μ IU/ml)	0.40–5.00	2.58

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for cholesterol to millimoles per liter, multiply by 0.02586.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

Treatment with intravenous furosemide was continued, but decongestive therapy was limited by low blood pressure, and the dose of metoprolol was reduced. Testing was negative for human immunodeficiency virus, hepatitis B and C viruses, and syphilis. The glycated hemoglobin level was 5.0% (reference range, 4.3 to 5.7). A test for anti-nuclear antibodies was positive at a titer of 1:40 (reference value, <1:40), and serum complement levels were within normal limits. An interferon- γ

release assay for *Mycobacterium tuberculosis* was negative. Serum protein electrophoresis and immunofixation showed an elevated IgA level of 829 mg per deciliter (reference range, 69 to 309), a low IgG level of 544 mg per deciliter (reference range, 614 to 1295), and a normal IgM level. Levels of serum free light chains were elevated, with a kappa light-chain level of 48.5 mg per liter (reference range, 3.3 to 19.4) and a lambda light-chain level of 101.6 mg per liter (reference range, 5.7 to 26.3). The serum free light-chain ratio (ratio of kappa to lambda free light chains) was normal at 0.48 (reference range, 0.30 to 1.70). Two IgA lambda M components (0.31 g per liter and 0.44 g per liter) were noted.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Zahir Kanjee: This 82-year-old woman presented to the emergency department of this hospital after a trip to South America, with obvious signs of volume overload on physical examination and imaging studies. Given that anasarca is the most distinct clinical feature of this patient's presentation, I will build the differential diagnosis around this finding.

ANASARCA

A physiological approach can be taken in determining the cause of anasarca. According to the Starling principle, edema occurs as a result of abnormal membrane permeability, an imbalance of oncotic pressure, elevated hydrostatic pressure inside a vessel, or a combination of these factors. Given how easily these diagnoses are overlooked, I will first consider anasarca associated with abnormal membrane permeability.

ABNORMAL MEMBRANE PERMEABILITY

Common causes of abnormal membrane permeability include angioedema, sepsis, burns, use of certain medications,¹ and pancreatitis. None of these conditions were present in this patient. Given the subacute and progressive nature of this patient's presentation, we can rule out idiopathic edema, which tends to occur in premenopausal women in a cyclic pattern, and systemic capillary leak syndrome, which has a more acute presentation and is associated with hypotension, hemoconcentration, and decreased jugular venous pressure, none of which were present initially in this patient.

IMBALANCE OF ONCOTIC PRESSURE

Lymphedema and myxedema are caused, in part, by increasing interstitial oncotic pressure. The distribution of the edema in this patient argues against lymphedema, and the normal thyrotropin level argues against myxedema. Her profoundly low blood level of albumin leads me to strongly consider decreased oncotic pressure as the cause of her anasarca. Any condition that leads to low albumin production, such as liver disease or malnutrition, or leads to additional loss of albumin, such as nephrotic syndrome or protein-losing enteropathies, should be ruled out. The patient does not appear to have overt malnutrition or gastrointestinal symptoms; thus, neither malnutrition nor a protein-losing enteropathy is a likely explanation for her condition.

INCREASED HYDROSTATIC PRESSURE

Common causes of increased hydrostatic pressure include heart failure and renal failure. Constrictive pericarditis is an easily overlooked diagnosis that leads to volume overload. Although Kussmaul's sign (paradoxical elevation of jugular venous pressure on inspiration) is easy to detect at the bedside, it is not uniformly present in patients with constriction.² Tuberculosis, which is common in South America, is a major cause of constriction globally. However, the absence of a clinically significant pericardial effusion, calcification, and scarring seen on nondedicated imaging studies renders constriction an unlikely diagnosis in this case. At this juncture, the most likely causes of the patient's anasarca include congestive heart failure, cirrhosis, and nephrotic syndrome, which I will discuss in detail.

CONGESTIVE HEART FAILURE

It is understandable that acute decompensated heart failure was the leading diagnosis upon this patient's admission, given her underlying history of heart failure with preserved ejection fraction. However, several aspects of her presentation are not consistent with this diagnosis. First, no apparent factors that would precipitate decompensated heart failure were present. Although many patients presenting with decompensation have no obvious cause, the absence of precipitating factors is nonetheless noteworthy. Furthermore, hypotension after initiation of intravenous diuresis and continuation of metoprolol would be unexpected in patients with typical heart failure

with preserved ejection fraction. Therefore, I am hesitant to attribute her presentation to an exacerbation of heart failure with preserved ejection fraction without considering a superimposed process, such as right ventricular failure, an ischemic cardiomyopathy with reduced ejection fraction, or an infiltrative cardiomyopathy such as amyloidosis. I will assume her murmur is preexisting and from tricuspid regurgitation; otherwise, I would want to investigate for primary right ventricular failure. It is important to note that primary cardiac causes would not explain the proteinuria or severe hypoalbuminemia in this patient. As such, we are left to consider other, extracardiac processes.

CIRRHOSIS

Cirrhosis should be considered in this patient, given that she presented with obesity and hypoalbuminemia, results of liver-function tests were abnormal, and the liver had a nodular contour on imaging. However, several features of her presentation argue against cirrhosis as the primary driver of her anasarca. Apart from the anasarca, no signs of chronic liver disease were present on physical examination, such as those caused by hyperestrogenism or portal hypertension. Although these features are more useful in establishing a diagnosis of cirrhosis than ruling it out,² the absence of any such findings is clinically significant. Furthermore, the normal results of coagulation tests and platelet count observed in this patient, as well as the absence of splenomegaly, are less consistent with cirrhosis.

NEPHROTIC SYNDROME

Nephrotic syndrome, defined by severe hypoalbuminemia and heavy proteinuria (urinary protein excretion of ≥ 3.5 g per day or protein:creatinine ratio [with protein measured in milligrams and creatinine in grams] of ≥ 3000), is often but not always accompanied by edema and hyperlipidemia.³ At first glance, it does not seem that there are sufficient data to make this diagnosis, but the patient also has several features consistent with nephrotic syndrome.

Two notable features of this patient's presentation are the presence of periorbital edema and frothy urine. Proteinuria is thought to lead to frothy urine by acting as a surfactant.⁴ In fact, the connection between frothy urine and kidney dis-

ease has been known since the time of Hippocrates, who suspected kidney disease to be the cause of anasarca in the presence of this finding.⁵ Although urine dipstick testing showed only 2+ proteinuria, this semiquantitative measurement does not rule out the presence of a large amount of albumin. Moreover, the dilute urine (based on the low specific gravity) further increases the possibility of heavy proteinuria. For all these reasons, I suspect that there is clinically significant proteinuria and that nephrotic syndrome is the most likely diagnosis in this patient.

On the basis of this assumption, I will turn to the differential diagnosis of nephrotic syndrome,⁵ which can be caused by several primary histologic conditions, such as minimal change disease, membranous nephropathy, focal segmental glomerulosclerosis, IgA nephropathy, and membranoproliferative glomerulonephritis. Many secondary causes of these conditions have already been ruled out by diagnostic testing. Since the patient was born in South America and had recently traveled there, parasitic diseases that can lead to various glomerular disorders causing proteinuria and nephrotic syndrome, such as Chagas' disease, filariasis, leishmaniasis, malaria, and schistosomiasis, should be considered.⁶ Her normal electrocardiogram argues against chronic Chagas' cardiomyopathy,⁷ and not many obvious or pathognomonic features of these other diseases were present. Hearing more about her specific geographic and exposure history would help refine my level of concern. However, subtypes of systemic amyloidosis that involve the kidneys and potentially the heart merit consideration — specifically, amyloid light-chain (AL) amyloidosis.

AL AMYLOIDOSIS

In addition to the suspected nephrotic syndrome, this patient has several features consistent with AL amyloidosis, including adverse reactions associated with cardiovascular medications, atrial fibrillation, results of liver tests showing a cholestatic pattern potentially consistent with infiltrative hepatopathy, clonal gammopathy, and elevated levels of light chains on serum testing. Cardiac amyloidosis is an increasingly recognized cause of chronic heart failure with preserved ejection fraction⁸; however, it is most often caused by wild-type transthyretin amyloidosis, rather than

AL amyloidosis. AL cardiomyopathy tends to be a more devastating and subacute process than that seen in this patient's history of chronic heart failure. Although her presentation does not include classic findings of AL amyloidosis, such as macroglossia, bruising, periorbital hemorrhage, peripheral neuropathy, and low QRS voltage on electrocardiography, these features are not always present. Her serum free light-chain ratio was normal; however, a small proportion of patients with AL amyloidosis may have a normal ratio. Despite the fact that AL amyloidosis is a rare entity, the preponderance of suspicious features makes this the most likely diagnosis.

To establish the diagnosis of AL amyloidosis, I would pursue a sequential workup, starting with the least invasive approach. First, I would confirm the diagnosis of nephrotic syndrome by measuring the spot urinary protein:creatinine ratio. Serologic testing for phospholipase A2 receptor antibodies is indicated to rule out primary membranous nephropathy and could obviate the need for a biopsy of the kidney. However, I suspect this patient will ultimately need a kidney biopsy, which could confirm a diagnosis of AL amyloidosis or any of several other possible causes of nephrotic syndrome.

DR. ZAHIR KANJEE'S DIAGNOSIS

Nephrotic syndrome, most likely due to AL amyloidosis.

HOME HOSPITAL COURSE

Dr. Travis: The ratios of total protein to creatinine and albumin to creatinine were elevated, with an estimated daily urinary excretion of 4.53 g of total protein (reference value, <0.15) and 3.26 g of total albumin (reference value, <30). A serum test was negative for antibodies against the M-type phospholipase A2 receptor. Antineutrophil cytoplasmic antibody (ANCA) testing was performed; immunofluorescence was weakly positive with an atypical nuclear pattern, and a positive enzyme-linked immunosorbent assay showed antibodies against proteinase 3 at a level of 43 U (reference value, negative). A diagnosis of nephrotic syndrome was made. At discharge from the home hospital service, the patient was referred to the

nephrology clinic of this hospital for evaluation of nephrotic syndrome.

NEPHROLOGY EVALUATION

Dr. Pui W. Cheung: The patient presented to the nephrology clinic with fluid overload and was noted to have edema involving her eyes, hands, legs, and abdominal wall. On the basis of a urinary protein:creatinine ratio of greater than 3.5 (both measured in grams), a serum albumin level of less than 2.5 g per deciliter, and the presence of diffuse anasarca, a diagnosis of nephrotic syndrome was made. Comprehensive serologic studies were performed, which ruled out most other common causes of nephrotic syndrome, such as systemic lupus erythematosus, diabetes mellitus, and primary membranous nephropathy. Although the ANCA titer was weakly positive, it is rare for ANCA-associated vasculitis to manifest as nephrotic syndrome without microscopic hematuria, which makes ANCA-associated vasculitis an unlikely diagnosis in this patient.

The serum free light-chain ratio in patients with normal kidney function ranges from 0.26 to 1.65.⁹ This ratio represents the balance between two types of light chains produced by plasma cells in the bone marrow and the rate of renal clearance. Therefore, this ratio is affected by kidney function, with the ratio increasing when kidney function is reduced because of impaired kappa light-chain filtration. Thus, a modified reference range of 0.82 to 3.60 has been established¹⁰ for the serum free light-chain ratio in patients with chronic kidney disease who have an estimated glomerular filtration rate (eGFR) of less than 55 ml per minute per 1.73 m². Further stratified reference ranges for the serum free light-chain ratio have been defined according to the degree of chronic kidney disease¹¹ in order to improve specificity and sensitivity for screening and monitoring of plasma-cell dyscrasias. Although the creatinine level was normal in this patient, the cystatin C–based eGFR was 34 to 41 ml per minute per 1.73 m², a finding suggestive of stage 3b chronic kidney disease. Even with the new reference ranges, establishing the diagnosis of AL amyloidosis in patients with chronic kidney disease on the basis of serologic tests alone remains challenging when the serum free light-chain ratio

falls on the edges of normal, as was seen in this patient.

Given the combination of clinical symptoms in this patient, including heart failure with hypotension and nephrotic syndrome, a high index of suspicion for systemic AL amyloidosis should be maintained, and a biopsy is warranted. A fat-pad biopsy was initially considered; however, owing to the initial weakly positive ANCA test, we cautiously proceeded with a biopsy of the kidney to rule out potential overlapping glomerulonephritis.

PATHOLOGICAL DISCUSSION

Dr. Lauren J. Ray: A core-needle biopsy of the kidney was performed (Fig. 2). Examination under light microscopy revealed 13 glomeruli, 15% of which were globally sclerosed, and mild tubular atrophy with interstitial fibrosis involving 15% of the cortex. Further inspection of the glomeruli showed mesangial expansion and capillary walls thickened by a nonargyrophilic, amorphous material, amyloid, which was mildly positive on periodic acid–Schiff staining and pale blue on Masson's trichrome staining. Subsequently, on Congo red staining, the mesangial and vessel wall material was salmon-pink under standard light microscopy with apple-green birefringence under polarized light.

Immunofluorescence studies performed on the tissue specimen showed broad amorphous deposits in the mesangium that were positive for lambda light chains, but staining for kappa light chains was negative. In addition, the glomeruli showed no pathologically significant staining for IgG, IgA, C3, C1q, or fibrin.

Electron microscopy was notable for randomly arranged fibrils, with a mean ($\pm$ SD) diameter of 10.5 ± 1.1 nm, within the mesangial and glomerular basement membrane deposits. Although the endothelial cells were largely preserved, podocytes had widespread effacement of foot processes associated with swelling and microvillous transformation.

Taken together, these features are diagnostic of AL amyloidosis (lambda light chain–type).

PATHOLOGICAL DIAGNOSIS

AL amyloidosis (lambda light chain–type).

NEPHROLOGY MANAGEMENT

Dr. Cheung: Managing nephrotic syndrome involves a multifaceted approach that focuses on addressing the underlying cause and managing the fluid overload. In addition to salt and fluid restriction, administration of diuretic and antiproteinuric agents is the mainstay of treatment. However, patients with hypoalbuminemia are particularly prone to hemodynamic-mediated kidney injury and have adverse reactions to many medications due to hypotension. This patient was persistently hypotensive with symptoms suggestive of pulmonary edema; therefore, we focused on volume management. Treatment with torsemide was started owing to better bioavailability than furosemide. Fortunately, this patient had a good response to torsemide alone, and her renal function remained stable. Once we obtained the biopsy report that established the diagnosis of AL amyloidosis, she was referred to the hematology service for further treatment.

ONCOLOGY MANAGEMENT

Dr. Andrew J. Yee: AL amyloidosis is a clonal plasma-cell abnormality. Other disorders that fall within this group are multiple myeloma and monoclonal gammopathy of undetermined significance (Fig. 3). What distinguishes AL amyloidosis from these conditions is the formation of amyloid deposits that lead to organ dysfunction. Occasionally, multiple myeloma can develop in patients with AL amyloidosis, and vice versa.

Several points of this patient's presentation merit highlighting. First, involvement of more than one organ is typical in patients with AL amyloidosis.¹² Second, the combination of serum and urine immunofixation electrophoresis and measurement of the free light-chain ratio is 99.8% sensitive for the diagnosis of AL amyloidosis.¹² However, it should be noted that 9% of these patients have a normal serum free light-chain ratio. Moreover, these tests are sensitive for the diagnosis of AL amyloidosis but are not specific; the majority of patients with abnormal laboratory findings on any one of these tests will receive a diagnosis of monoclonal gammopathy of undetermined significance, which is a far more common condition.

Since AL amyloidosis and multiple myeloma are

both plasma-cell disorders, their treatment strategies share common elements, such as the incorporation of a proteasome inhibitor and, more recently, the anti-CD38 monoclonal antibody daratumumab. By treating the source of the amyloid fibrils, therapies that target the clonal plasma cell can decrease the amyloidogenic light-chain burden and improve organ function. The ANDROMEDA study showed that the addition of daratumumab to the combination therapy of bortezomib, cyclophosphamide, and dexamethasone led to significant improvements in overall survival and 5-year progression-free survival, and it has been shown to decrease proteinuria.¹³

BONE MARROW BIOPSY

Dr. Ray: A bone marrow biopsy was performed (Fig. 4). Examination of the biopsy specimen under light microscopy showed normocellular marrow with maturing trilineage hematopoiesis. Giemsa staining and immunohistochemical staining for CD138 showed small, mature-appearing plasma cells that comprised 5 to 10% of the cellularity. Analysis of kappa and lambda light-chain expression with the use of in situ hybridization confirmed monoclonality of the plasma cells with lambda light-chain restriction. The periosteal soft tissue contained numerous small vessels with walls thickened by an amorphous material that appeared salmon-pink on Congo red staining and showed apple-green birefringence under polarized light.

Flow cytometry of the bone marrow specimen showed a monoclonic lambda plasma-cell population (0.82% total events) with expression of CD38 and CD138. Fluorescence in situ hybridization revealed an *FGFR3-IGH* fusion, which usually represents t(4;14) translocation. A targeted sequencing panel for 88 genes that are commonly altered in hematologic cancers was performed and showed no pathogenic alterations.

Overall, these findings are consistent with a plasma-cell neoplasm resulting in immunoglobulin-related amyloidosis (AL amyloidosis).

FOLLOW-UP

Dr. Yee: Given the patient's older age and frailty, treatment with single-agent daratumumab was administered, along with dexamethasone. Monotherapy with daratumumab has an excellent bal-

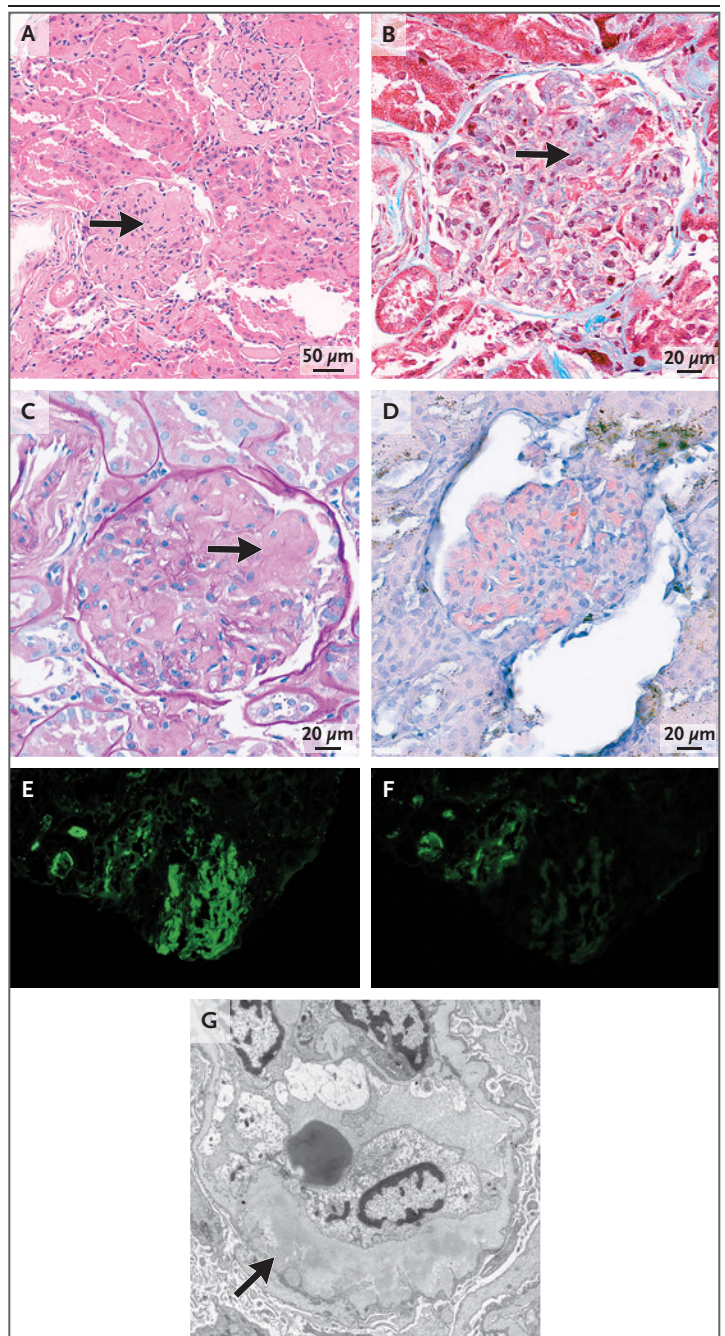
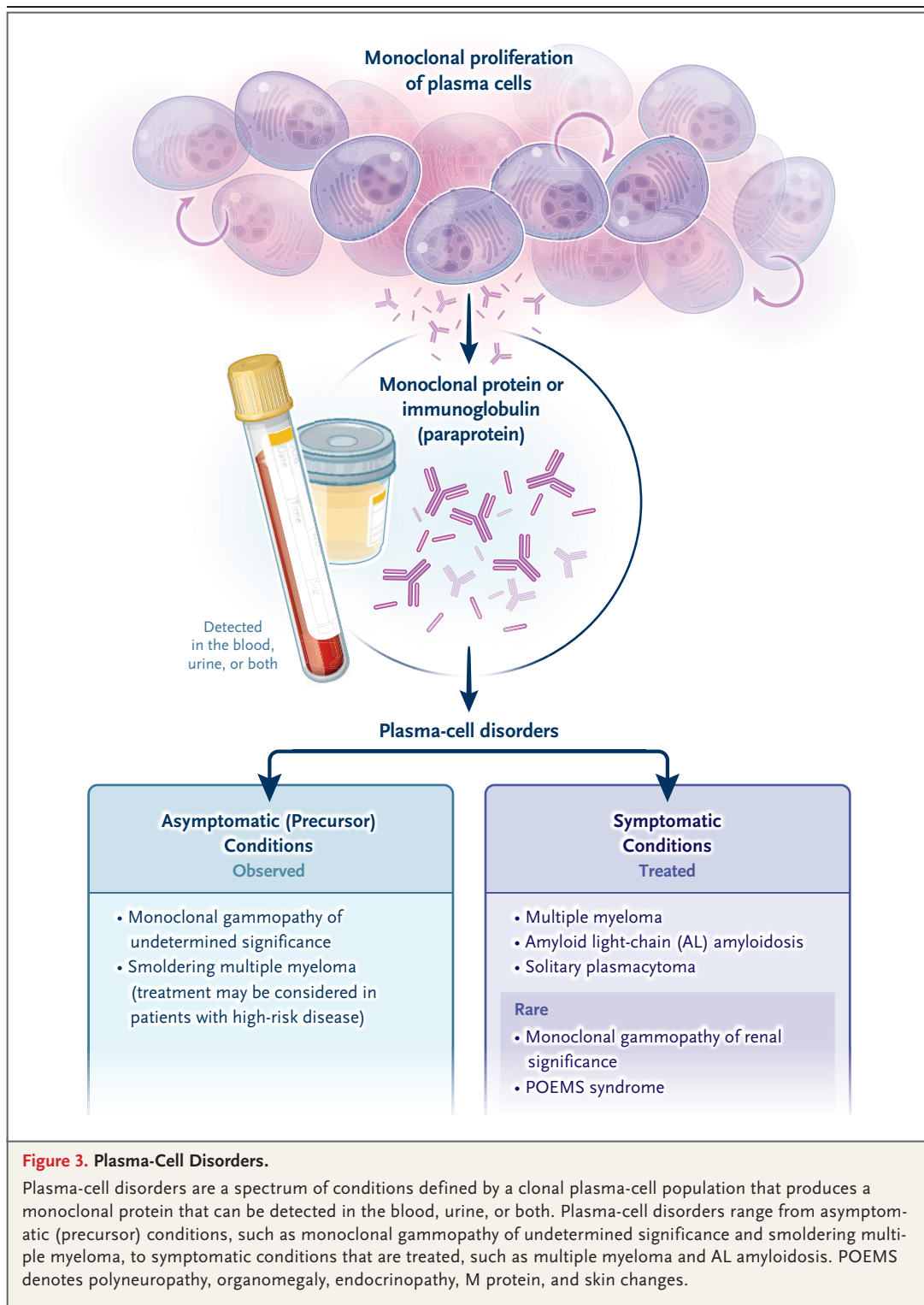


Figure 2. Biopsy Specimen of the Kidney.

Hematoxylin and eosin staining shows mesangial expansion by an amorphous material (Panel A, arrow) consistent with amyloid is seen in the glomeruli. The amyloid has a characteristic pale blue appearance on Masson's trichrome staining (Panel B, arrow) and is mildly positive on periodic acid–Schiff staining (Panel C, arrow). The amyloid within the mesangium has a salmon-pink appearance on Congo red staining (Panel D). Immunofluorescence of the amyloid deposits shows lambda light-chain restriction (Panel E) and is negative for kappa light chains (Panel F). On electron microscopy, amyloid deposition is seen within the glomerular basement membrane (Panel G, arrow).



ance of efficacy and safety in patients with multisystem AL amyloidosis.¹⁴ At a follow-up visit 1 year after her initial presentation, the patient was

feeling well, and the edema in her lower legs had resolved. The difference between the serum free kappa and free lambda light chains had decreased

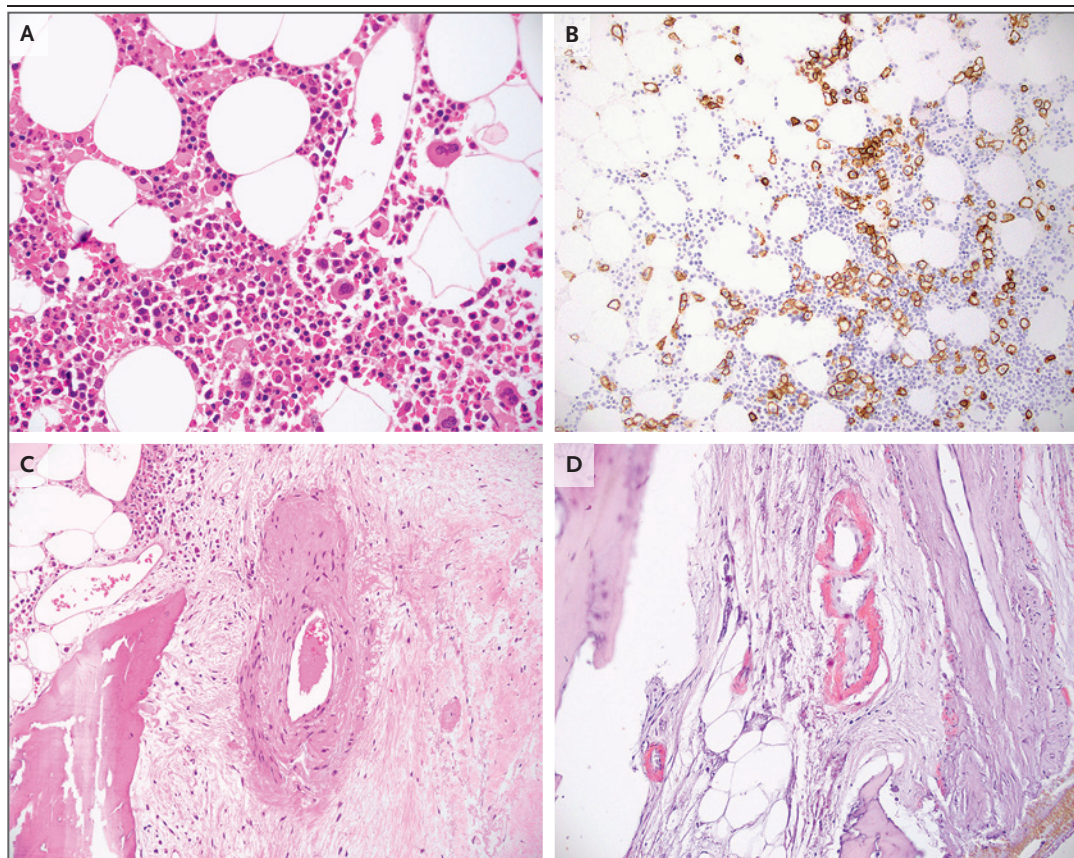


Figure 4. Bone Marrow Biopsy Specimen.

Hematoxylin and eosin staining of a bone marrow biopsy specimen (Panel A) shows normocellular marrow with small clusters of plasma cells, which are further highlighted on immunohistochemical staining for CD138 (Panel B). Hematoxylin and eosin staining of the periosteal soft tissue (Panel C) shows amyloid deposition within the vessel walls. On Congo red staining (Panel D), these deposits are salmon-pink under standard light microscopy and showed apple-green birefringence under polarized light (not shown).

from 53.1 mg per liter to 0.4 mg per liter, and proteinuria, as measured by the ratio of spot protein to creatinine in urine, had also decreased from 4.53 to 0.65 g of total protein per day.

FINAL DIAGNOSIS

AL amyloidosis.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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